

■ 2.1 Expressions

■ 2.1.1 Everything Is an Expression

Mathematica handles many different kinds of things: mathematical formulae, lists, graphs, to name a few. Although they often look very different, *Mathematica* represents all of these things in one uniform way. They are all *expressions*.

A prototypical example of an *Mathematica* expression is `f[x, y]`. You might use `f[x, y]` to represent a mathematical function $f(x, y)$. The function is named `f`, and has two arguments, `x` and `y`.

You do not always have to write expressions in the form `f[x, y, ...]`. For example, `x + y` is also an expression. When you type in `x + y`, *Mathematica* converts it to the standard form `Plus[x, y]`. Then, when it prints it out again, it gives it as `x + y`.

The same is true of other “operators”, such as `^` (**Power**) and `/` (**Divide**).

In fact, everything you type into *Mathematica* is treated as an expression.

<code>x + y + z</code>	<code>Plus[x, y, z]</code>
<code>x y z</code>	<code>Times[x, y, z]</code>
<code>x^n</code>	<code>Power[x, n]</code>
<code>{a, b, c}</code>	<code>List[a, b, c]</code>
<code>a -> b</code>	<code>Rule[a, b]</code>
<code>a = b</code>	<code>Set[a, b]</code>

Some examples of *Mathematica* expressions.

You can see the full form of any expression by using `FullForm[expr]`.

Here is an expression.

```
In[1]:= x + y + z
Out[1]= x + y + z
```

This is the full form of the expression.

```
In[2]:= FullForm[%]
Out[2]//FullForm= Plus[x, y, z]
```

Here is another expression.

```
In[3]:= 1 + x^2 + (y + z)^2
```

```
Out[3]= 1 + x2 + (y + z)2
```

Its full form has several nested pieces.

```
In[4]:= FullForm[%]
```

```
Out[4]//FullForm=
```

```
Plus[1, Power[x, 2], Power[Plus[y, z], 2]]
```

The object f in an expression $f[x, y, \dots]$ is known as the *head* of the expression. You can extract it using `Head[expr]`. Particularly when you write programs in *Mathematica*, you will often want to test the head of an expression to find out what kind of thing the expression is.

Head gives the “function name” f .

```
In[5]:= Head[f[x, y]]
```

```
Out[5]= f
```

Here Head gives the name of the “operator”.

```
In[6]:= Head[a + b + c]
```

```
Out[6]= Plus
```

Everything has a head.

```
In[7]:= Head[{a, b, c}]
```

```
Out[7]= List
```

Numbers also have heads.

```
In[8]:= Head[23432]
```

```
Out[8]= Integer
```

You can distinguish different kinds of numbers by their heads.

```
In[9]:= Head[345.6]
```

```
Out[9]= Real
```

<code>Head[expr]</code>	give the head of an expression: the f in $f[x, y]$
<code>FullForm[expr]</code>	display an expression in the full form used by <i>Mathematica</i>

Functions for manipulating expressions.